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Space | North America

# Varda Space: Space Manufacturing, Pharma Made in Orbit

One day, your doctor's prescription may have been produced in the microgravity of space. The benefits of manufacturing complex compounds in space have been documented for decades, but never commercialized.



Please note that we are reposting due to earlier technical issues with file size.

## Is the Space Economy about to get a pharmaceutical sector?

- Today, the space economy is commonly framed around two dominant verticals (launch vehicles/infrastructure and satellite communications) which have attracted the majority of capital, attention, and commercial activity to date.
- However, the range of commercial opportunities available is much wider, with the expansion of those foundational industries setting the necessary infrastructure for a surge of new opportunities ranging from manufacturing, defense, mining, compute, power, exploration, and more.
- The concept of manufacturing in space is not new, dating back to the early 1970's [NASA Skylab](#) experiments. But it has never been commercialized at scale given limited launch cadence, launch costs, and regulatory complexities, among other reasons.
- Why manufacture in space? Unique product potential (eliminate gravitational disturbances), reducing reliance on launch, in-situ resource utilization, etc. It's difficult to get more 'first-principles' than altering the laws of physics.
- In medicine, [scientists have proposed that](#) by improving the size, uniformity, and purity of the active pharmaceutical ingredients (APIs), companies can theoretically enable new drugs or drug delivery options.
- In 1998, [insulin crystals grown on Space Shuttle Mission STS-95](#) were shown to be unusually large: on average 34x larger, with 7x lower mosaicity (imperfections), and 54x higher reflection peak heights than Earth-grown crystals.
- Reusable rockets are an effective elevator to space, having thus far reduced launch costs/kg by a factor of ~10x. With a new generation of larger, fully reusable vehicles under development, launch economics may continue to meaningfully improve over the long-term, further expanding the range of commercially viable activities in space.
- Will it be challenging? Obviously. Regulatory barriers, process control,

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uncertain economics, high-volume launch demand, etc. all pose meaningful hurdles. There is a reason the technology has taken decades to become commercialized in any capacity.

- *In this note, we profile Varda Space, a company that designs and flies spacecraft to manufacture existing pharmaceuticals in low Earth orbit, leveraging the microgravity of space to create drug molecules with purities unachievable on Earth - potentially allowing pharma companies to improve shelf life, bioavailability, and even enable new methods of drug delivery to improve patient experience.*

**Varda/In-Space Pharma Manufacturing Explained:** Most medicines, especially pills, are made from tiny crystals of the drug, and the shape and quality of these crystals affect how well the medicine dissolves, absorbs, stays stable, and can be manufactured. In space, the microgravity environment lets crystals grow more evenly and symmetrically, with fewer defects, making them purer and more uniform than what's typically possible on Earth. Companies like Varda Space send small, automated factories into orbit to grow these higher quality crystals and then bring them back to Earth. With these improved crystals, a good but flawed drug can potentially become a great one, a drug than previously failed in development might become usable, and a hard to manufacture drug may be produced at scale.

**Exhibit 1:** Varda Space Company Profile



**FOUNDERS**

|                                                                                                                                                                    |                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>Will Bruey</b><br>Co-Founder, CEO<br>Prior:<br>SpaceX<br>Bank of America | <br><b>Delian Asparouhov</b><br>Co-Founder, President<br>Prior/Others:<br>Founders Fund<br>Khosla Ventures<br>Nightingale App |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Notable Investors**  
\$328mm Funding Raised (Valuation Unknown)

|                  |                     |
|------------------|---------------------|
| FOUNDERS FUND    | khosla ventures     |
| LU+              | CAFFEINATED CAPITAL |
| AlsoCapital+     | Alumni Ventures     |
| SHRUG            | NATURAL CAPITAL     |
| GENERAL CATALYST |                     |

Morgan Stanley

Source: Varda Space, NASA, PitchBook, Morgan Stanley Research

**For Further Reading:**

- [Physical AI: The Morgan Stanley Robot Almanac \(7 Dec 2025\)](#)
- [Space: Starcloud: Data Centers in Space - AI Goes Orbital \(27 Jan 2026\)](#)
- [Video: Autonomy at Altitude: Joby Aviation \(22 Jan 2026\)](#)
- [Rowdy Robot: OpenAI RFP Aligns with Rebuilding American Manufacturing](#)

[Muscle \(22 Jan 2026\)](#)

- [Defense Tech: Anduril: a NeoPrime\(r\) \(12 Jan 2026\)](#)
- [Embodied AI: Saronic: Robots Ahoy! \(We're Gonna Need a Better Boat\) \(2 Nov 2025\)](#)
- [Robotics: I Just Pre-Ordered a Humanoid Robot \(4 Nov 2025\)](#)
- [Embodied AI: Humanoid Disruptors: Unitree - Humanoids Starting at \\$16K \(28 Jan 2025\)](#)
- [Embodied AI: AI Robotics Disruptors: Physical Intelligence \(19 Dec 2024\)](#)

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# About Varda Space & In-Space Manufacturing

**Founding Story:** [Varda Space Industries](#) was officially founded in January 2021 to commercialize in-space manufacturing, based on the premise that microgravity enables improved material and pharmaceutical processes, particularly crystal growth, that are difficult to achieve consistently on Earth. The company was co-founded by [Will Bruey](#), a former SpaceX engineer who was responsible for avionics design and mission control of the Dragon spacecraft, and [Delian Asparouhov](#), a Founders Fund partner who [previously pursued the thesis](#) that falling launch costs could make orbital manufacturing economically viable at scale, but realized there was a lack of companies to invest in.

From inception, Varda focused on building free-flying spacecraft capable of both performing manufacturing in orbit and returning finished product to Earth, viewing reentry and recovery as the critical bottleneck to commercialization. The company raised [seed funding in late 2020](#) from Founders Fund and Lux Capital, enabling early vehicle design and team formation ahead of formal incorporation. In its early years, Varda prioritized speed to flight, contracting satellite bus capacity from Rocket Lab (RKL - Covered by Kristine Liwag) while developing a proprietary reentry capsule leveraging novel heatshield material from NASA. This approach eventually culminated in Varda's [W-1 mission](#), which successfully manufactured pharmaceutical crystals of the HIV drug, Ritonavir, in orbit and returned them to Earth in early 2024, marking a key proof point for the business model. Today, Varda positions itself as an emerging space-based manufacturing company aiming to provide repeatable, returnable microgravity production for high-value materials, with production of active pharmaceutical ingredients (API's) as the initial focus.

**Exhibit 2:** Varda Space Founders Delian Asparouhov and Will Bruey



Source: Varda Space

**What is In-Space Manufacturing?** [In-Space Manufacturing \(ISM\)](#) refers to producing, processing, or assembling materials and products in the space environment, most commonly in low Earth orbit, where sustained microgravity changes fluid flow, heat transfer, sedimentation, and crystal growth. The objective is typically either to leverage improved material properties versus Earth-based production (Space-for-Earth manufacturing), or to leverage the ability to produce and assemble complex structures for

use in space to mitigate launch costs and related complexities (Space-for-Space manufacturing)

*The idea is not new and has existed since the early days of space exploration.* Microgravity materials processing and crystal growth concepts were explored as early as the [Skylab mission \(1973\)](#), with work later expanded on through Space Shuttle missions, Mir, and on the ISS, where a steady cadence of materials science and life science research helped identify “production-adjacent” use cases. Commercial activity has gradually expanded from astronaut-tended experiments to more automated, product-oriented demonstrations, including [fiber optics](#) work and other advanced materials.

In parallel, some companies are pursuing “factory-to-return” logistics, where a free-flying spacecraft performs processing in orbit and returns the payload to Earth. Varda Space is one example of this direction, emphasizing a recurring orbital platform coupled with a reentry capsule to close the loop from in-orbit processing to terrestrial delivery.

- **Key Rationale:**

- **Reusable rockets = elevators to space.** Reusable rockets have thus far reduced launch costs/kg by a factor of ~10x on our estimates (Falcon 9 priced at ~\$2-4k per kg) with new reusable heavy and super-heavy lift rockets (Starship, Blue Glenn, etc.) likely to reduce launch costs by a further magnitude. We would expect that continued significant declines in launch costs to gradually enable a range of space-based industries and infrastructure that were previously un-economical.
- **Unique product quality potential.** Certain materials exhibit properties in microgravity that are difficult or impossible to replicate on Earth. For example, optical fiber experiments on the ISS aim to produce higher-quality glass with fewer gravity-induced defects versus terrestrial processes, and companies have demonstrated ZBLAN fiber pulls in orbit. In medicine (where Varda focuses), [insulin crystals grown on Space Shuttle Mission STS-95](#) were shown to be unusually large- on average 34x larger with 54x higher reflection peak heights than Earth-grown crystals.
- **Enabling build out of space infrastructure.** On-orbit production and assembly could reduce reliance on Earth launches for spacecraft components and infrastructure, lowering long-term logistics costs and enabling larger or more complex infrastructure systems. For example, imagine a spare part being needed on a space station. Instead of waiting for the next resupply launch, that component could theoretically be additively manufactured for in-orbit use.
- **Long-term exploration and ISRU linkage.** In situ resource utilization (ISRU) concepts aim to convert lunar, asteroid, or Martian materials into propellant, construction products, or life-support consumables, reducing Earth supply chain dependency for deep space missions.

- **Notable Challenges:**

- **Uncertain unit economics.** Launch, operations, and reentry costs still remain high on a per-kilogram basis, and the value density required to justify these costs is only proven for a few niche materials to date. The sector’s path to profitable scale remains unproven and may require a very large amount of

capital investment prior to meaningful revenue generation.

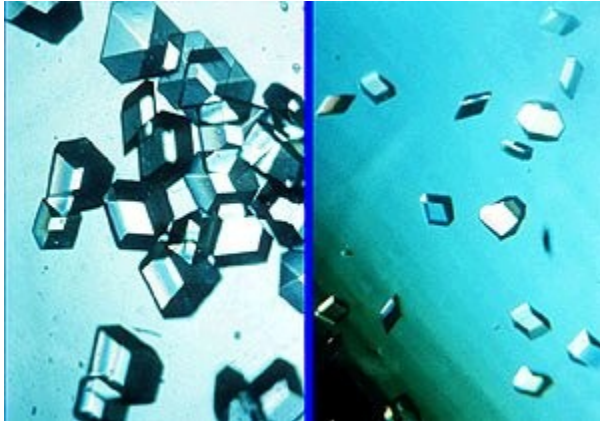
- **High volume launch demand.** In-space manufacturing companies like Varda effectively generate revenue by producing and returning a consumable output that is physically constrained to each spacecraft. Sustaining revenue therefore requires frequent, reliable launch and return cadence - tying commercial throughput directly to access-to-space capacity, scheduling availability, and vehicle reliability.
- **Process control and environmental complexity.** Microgravity, vacuum, temperature extremes, and radiation [can make process stability and reproducibility challenging](#) relative to terrestrial manufacturing. Even successful experiments require iterations to become industrially reliable.
- **Return and logistics engineering overhead.** For space-for-Earth products, returning goods requires thermal protection, tracking, recovery infrastructure, and regulatory coordination. Each mission adds integration complexity and schedule risk.
- **Limited proven near-term demand.** Despite recent interest from government (including the DoW) and commercial customers, buyers of made-in-space products are likely to require clear evidence that microgravity adds meaningful value, especially given the complexities and related costs.
- *For more information on ISM's rationale and challenges, we recommend reading the following 2025 article from the University of Nevada-Reno: [In-Space Manufacturing: Technologies, Challenges, and Future Horizons](#)*

**Current ISS Pharmaceutical Activity.** The International Space Station hosts ongoing microgravity life sciences research, including [protein crystal growth](#), [monoclonal antibody formulation](#), and drug stability studies. Through the ISS National Lab, pharmaceutical companies such as Merck and Eli Lilly have evaluated whether microgravity can improve crystal size, uniformity, or enable alternative polymorphs for existing drugs ([see recent Merck ISS research on its cancer drug, pembrolizumab](#) AKA 'Keytruda'). While this work remains largely at the R&D stage rather than scaled production, the ISS has served as a foundational proving ground for the scientific premise that microgravity can influence pharmaceutical structure and performance, an insight companies like Varda aim to translate into repeatable manufacturing.

- **Could Space Stations Compete with Varda?** Definitely. Commercial space stations ([Starlab](#), Vast, Axiom, etc.) could theoretically host pharmaceutical manufacturing payloads, offering continuous power and longer-duration microgravity access. However, this approach carries trade-offs Varda seeks to avoid, including higher fixed infrastructure costs, shared resource constraints, crew scheduling complexity, and additional complexities with de-orbiting manufactured pharmaceuticals. Varda CEO Will Bruey has stated in the past that while these companies could become competitors, they also present a potential partnership opportunity for the company.

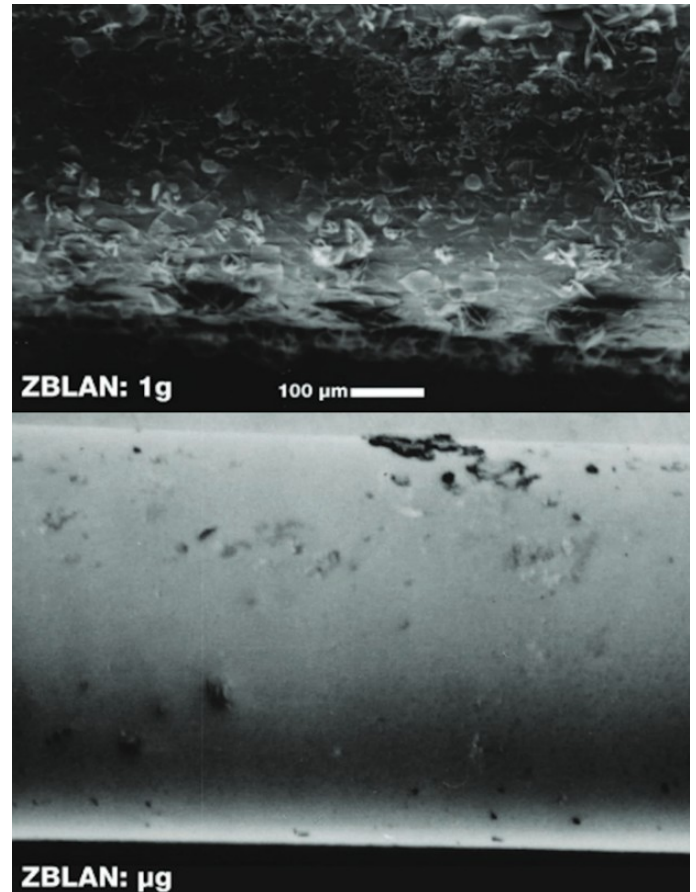


**Exhibit 3:** Insulin Crystals Grown in Space (left) and on Earth (right). Crystals grown on Space Shuttle Mission STS-95 were on average 34x larger than Earth-grown counterparts.



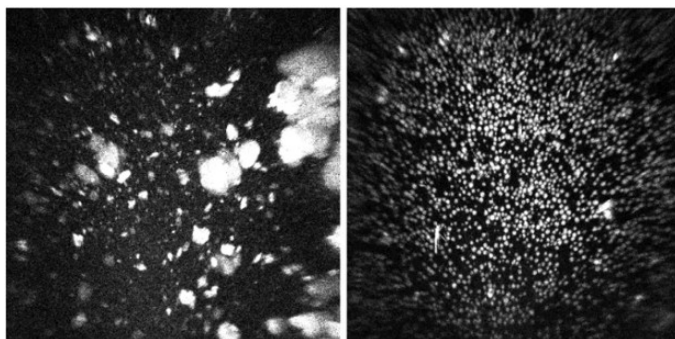
Source: NASA

**Exhibit 4:** ZBLAN Optical Fiber Pulled in Space (bottom) and on Earth (top)



Source: NASA

**Exhibit 5:** Pembrolizumab (cancer medicine) Crystals Grown on the ISS (right) vs. Ground Sample (left)



Source: Merck, NASA

**Varda's Business Model:** Operationally, Varda's model is to fly a contained pharmaceutical processing payload on a small uncrewed spacecraft capsule, run the crystallization/manufacturing process in orbit, and then return the product to Earth in a reentry capsule so it can be characterized and, if valuable, integrated into conventional downstream pharma development and manufacturing. The company does not develop drugs or help create new drug molecules. Instead, they are taking existing drugs manufactured by major pharma customers and allowing them to create versions with

increased purity and density.

Varda Space's CEO Will Bruey has indicated the company can theoretically produce other materials which benefit from microgravity in space (optical fibers, silicon wafers, etc.), but for the foreseeable future, given the highest value chemicals (price and human impact) are pharmaceuticals, the company is focused entirely on that end market.

**How Does In-Space Pharmaceutical Manufacturing Work?** Many of the active ingredients in drugs are produced and formulated as solid crystals grown from a liquid solution. On Earth, gravity-driven fluid motion (buoyancy-driven convection), sedimentation, and related mixing effects can disturb the growth environment around a crystal, influencing crystal size, defect density, and even which crystal form (polymorph) appears. In microgravity in Low Earth Orbit (LEO), [convection and sedimentation are greatly reduced](#), which can enable more ordered crystals with fewer defects and, in some cases, different crystal forms than are impractical to obtain on Earth.

- **How Are Drugs Chosen?** Varda's stated approach is to screen drug substances on Earth for gravity sensitivity using a [hypergravity crystallization platform \(a large centrifuge-based setup\)](#) that varies effective gravity to observe how changes in gravity affect nucleation and crystal growth behavior. The idea is to quickly identify candidates where gravity materially changes attributes like crystal habit, particle size distribution, or polymorph outcomes.
- **Why Complicate Drug Manufacturing?** By improving the size, uniformity, and purity of the active ingredients, pharma companies can theoretically enable new drugs or drug delivery options. For example, [recent research on the International Space Station with Merck](#) has suggested that microgravity crystallization could enable reformulations of some cancer drugs, theoretically turning lengthy intravenous infusions into simpler injections by producing crystals with characteristics more suitable for alternative delivery routes. Other drugs have high level of impurities which can become trapped within the crystal lattice structures when formed, creating risk of toxic side effects. However, the slower and more uniform crystallization of drugs in microgravity (all else equal) allows for an easier separation of impurities, theoretically resulting in a drug that is more pure and high density (example given by CEO Bruey in [podcast with Siemens](#)).
- **What Has Varda Demonstrated Thus Far?** To date, Varda has successfully demonstrated commercial in-space pharmaceutical manufacturing with multiple missions. Its [W-1 mission crystallized the HIV antiviral drug Ritonavir](#) in low Earth orbit and safely returned the metastable Form III crystals to Earth for analysis. Varda [collaborated with pharma research company, Improved Pharma](#), on post-flight analysis showing that Ritonavir crystals remained stable after atmospheric reentry demonstrating medicine survivability. In subsequent flights, Varda has continued to fly pharmaceutical payloads alongside other experiments on missions W-2 and W-4 for undisclosed customers/partners.

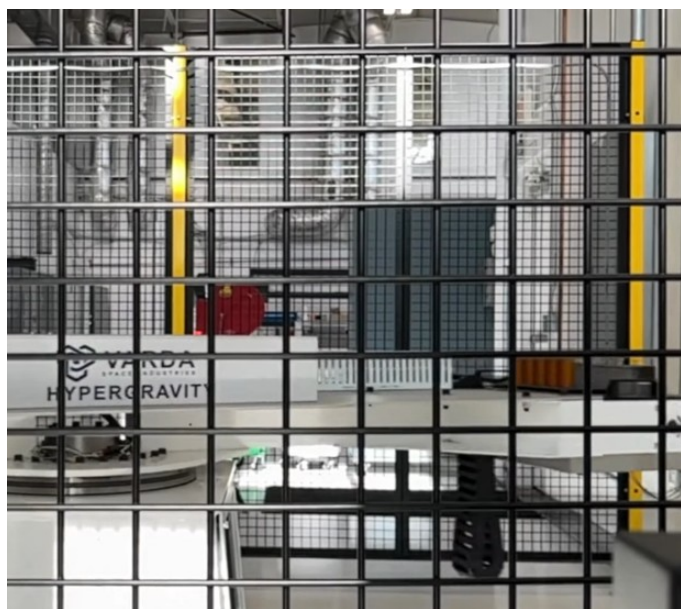
**FDA Approval - A Key Hurdle with Unsolved Challenges:** Pharma missions flown by Varda today have been largely technical demonstrations for safety and technological validation. However, to date, there are no completed regulatory approvals specifically for drugs manufactured in space for human use. Any such drug would likely need to go through the standard US Food and Drug Administration (FDA) approval pathways, such as



a New Drug Application (NDA) or equivalent route demonstrating safety and efficacy through clinical data, regardless of where the active ingredient was produced.

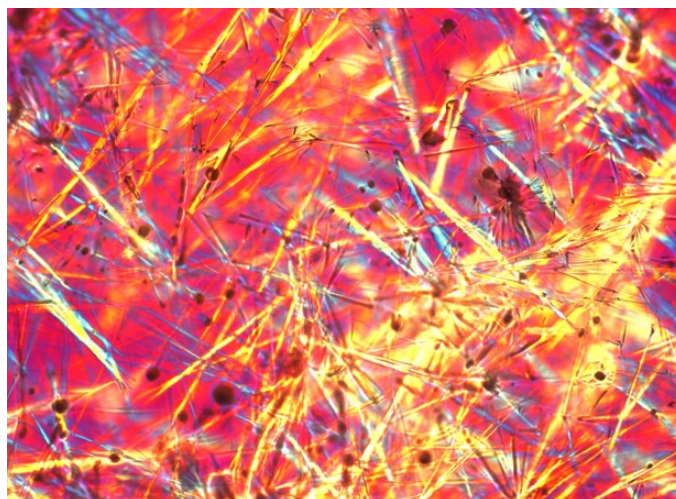
Currently, the FDA conducts [CGMP \(Current Good Manufacturing Practices\) inspections](#) of drug manufacturing facilities to ensure their safety, which is obviously complicated when these facilities are now placed in orbit. *It is an unsolved challenge*, with lawyers speculating that the FDA would need to rely on some form of [Remote Regulatory Assessment](#) for in-space manufacturing which were used extensively during the travel restrictions of the COVID-19 pandemic. Although, this would likely be more relevant to manufacturing on space stations that remain in orbit and re-used over a longer period of time, and since Varda's spacecraft is only in-orbit for a few weeks or months at a time, may require a completely novel approach.

**Exhibit 6:** Hypergravity Screening via Centrifuge for Potential In-Space Production



Source: Varda Space

**Exhibit 7:** Ritonavir (For HIV Treatment) Crystallization



Source: Varda Space

**W-Series (Winnebago) Spacecraft:** [Varda's W-Series](#) (AKA 'Winnebago') is free-flying orbital processing and return system designed to manufacture or process materials in low Earth orbit and then bring the output back to Earth in a dedicated reentry capsule. Today, Varda positions the platform primarily around microgravity-enabled pharmaceutical processing, while also marketing the system as a repeatable reentry logistics platform for government and hypersonic-related payloads.

- **Design Specs:** The spacecraft weighs ~300kg total, is designed for missions ranging from a few weeks to a few months, and is built to withstand reentry speeds exceeding Mach 25. The architecture pairs (1) a 1-meter diameter reentry capsule that houses the manufacturing payload and returns to land under parachute, with (2) an attached satellite bus that provides on-orbit power, communications, attitude control, and propulsion, including deorbit targeting prior to capsule separation.
- **Satellite Bus Design and Heat Shielding:** Missions W-1 through W-3 were based

on an off-the-shelf Rocket Lab Photon satellite bus while later missions leveraged a [Varda-designed satellite bus](#) designed as part of a NASA Tipping Point technology transfer agreement giving the company access to produce C-PICA, a thermal protection material developed at NASA's Ames Research Center for re-entry vehicles, at Varda's factories.

- **Payload Support:** Varda markets configurable payload support, citing over 100 W of power and tens of kilograms of internal mass capacity for payloads, alongside thermal protection area suitable for instrumentation and testing.
- **Why So Small?** The capsule is intentionally compact (about one meter in diameter) because it only manufactures the active pharmaceutical ingredient (API), which generally makes up a very small portion of the final drug delivered to patients. Producing APIs in space doesn't require large volumes to be commercially relevant. Smaller batches further allows Varda's pharma customers to refine the production through repeated iteration. Also, the small form factor also lets the spacecraft fly on Falcon 9 rideshare missions, which is important for keeping launch costs down and enabling a frequent flight cadence.
- **Flight Cadence:** Eventually, Varda hopes to operate numerous spacecraft simultaneously (W-4 and W-5 missions were first to operate at same time) with a longer-term goal of weekly or even daily flights at scale. The company has publicly [indicated](#) that it expects a near-monthly reentry cadence by end of 2028. The company's next mission, W-6 is tentatively planned for March.

**FAA Launch/Reentry Licensing:** We note that Varda was granted the first ever FAA Part 450 reentry license allowing its Winnebago capsules to reenter Earth's atmosphere and land, initially enabling the return of its [W-1 mission](#) in early 2024 and [later expanded](#) under a vehicle operator license through 2029 to permit routine reentries without filing separate safety documentation for each flight (first leveraged for W-4 mission).

**Exhibit 8:** Assembly of W-2 Spacecraft



Source: Varda Space

**Exhibit 9:** View of W-4 Spacecraft



Source: Varda Space

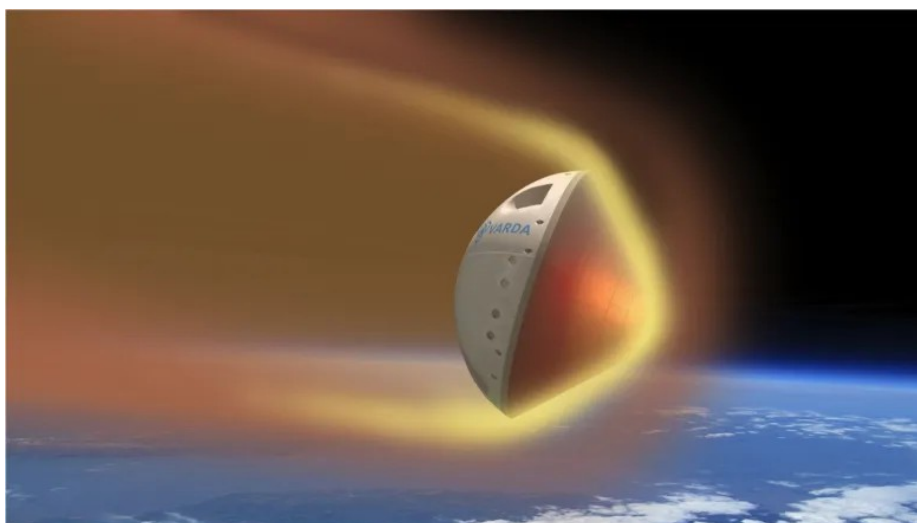
#### Industries Served:

- **Biopharma:** Biopharma is Varda's primary commercial focus, centered on microgravity-enabled crystallization of existing active pharmaceutical ingredients

(APIs). The company does not develop new drug molecules, but instead enables pharmaceutical companies to manufacture alternative crystal forms or higher-purity versions of known compounds that may be difficult to achieve on Earth. We note that current activity appears limited to R&D rather than making drugs intended to be administered to humans, and we are not aware of any current FDA approvals that would enable the company to do this at this time.

- **Government:** Varda positions its W-Series spacecraft not only as a manufacturing platform but also as a [orbital test and reentry vehicle for US government customers](#) to test hypersonic and reentry technology such as heatshields. Varda previously carried government payloads on W-2 (NASA heatshield, US Air Force spectrometer), W-3 (US Air Force IMU), and W-5 (US Navy hypersonic data collection)
- **Microgravity Research:** As a more open-ended market opportunity, Varda also serves the broader microgravity research community by offering a free-flying, automated alternative to ISS-based experiments, particularly for studies requiring autonomous operation and physical return of samples. An example of Varda executing on this opportunity would be the October 2025 multi-flight [partnership with United Semiconductors](#), a company developing in-space semiconductor 'foundries', to produce semis in orbit.

**Exhibit 10:** Varda also offers a testbed for US Government testing of hypersonic technology, having carried military payloads on W-2, W-3, and W-5



Source: Varda Space

#### Mission History:

- **W-1 (Jun '23 - Feb '24):** First Varda reentry mission aimed at demonstrating in-space pharmaceutical manufacturing by growing drug crystals ([Ritonavir](#)) in microgravity and returning them intact to Earth; capsule reentered and landed at the Utah Test and Training Range under FAA Part 450 approval. This flight marked the first commercial reentry of its kind and validated Varda's heatshield and reentry systems.
- **W-2 (Jan '25 - Feb '25):** Dual-purpose mission carrying Varda's orbital manufacturing payload plus a spectrometer from the U.S. Air Force Research



Laboratory (AFRL) and NASA-collaborated thermal protection hardware; launched on SpaceX Transporter-12 and returned to the Koonibba Test Range in Australia. The flight provided hypersonic reentry data and tested expanded payload capabilities beyond pure manufacturing.

- **W-3 (Mar '25 - May '25):** AFRL Prometheus program-funded testbed for U.S. Air Force hypersonic technology, flying an advanced inertial measurement unit (IMU) developed with USAF and ISSI; reentered at Koonibba after ~2 months on orbit. This was principally a national-security/hypersonics tech validation mission support rather than purely manufacturing.
- **W-4 (Jun '25 - Present):** Launched aboard Transporter-14; first mission using Varda's own in-house developed spacecraft bus and integrated systems rather than the Rocket Lab platform used on previous missions. The vehicle is hosting extended in-orbit manufacturing experiments (pharma and semiconductor processes) and represents a shift toward vertically integrated spacecraft production.
- **W-5 (Nov '25 - Jan '26):** Fourth launch of 2025, carrying a specialized U.S. Navy payload focused on data collection during atmospheric reentry; first use of Varda's vertically integrated satellite bus for a complete mission from orbital ops to capsule return. The spacecraft conducted a successful reentry on January 29, 2026.
- **W-6 (Scheduled Mar '26):** Upcoming W-series mission with limited public detail available; expected to continue high-cadence reentry operations and likely serve NASA, defense, or commercial partners under Varda's standard orbital processing and return model. (No detailed mission objectives publicly published as of current data.)

**Exhibit 11:** W-1 Following Landing at Utah Test and Training Range



Source: Varda Space

**Exhibit 12:** W-3 Capsule View During Reentry Procedures



Source: Varda Space

#### Partnerships/Suppliers:

- **SpaceX (Launch Services):** Varda has used SpaceX as its primary launch provider, leveraging Falcon 9 rideshare missions to deploy its autonomous orbital manufacturing capsules.
- **Rocket Lab (Spacecraft Platforms):** Early in its operations, Varda worked with Rocket Lab to [procure Photon/Pioneer satellite buses](#) and integrate spacecraft

systems for its W-Series orbital manufacturing missions. However, beginning with the W-4 mission in June 2025, Varda has transitioned to its own in-house developed satellite bus.

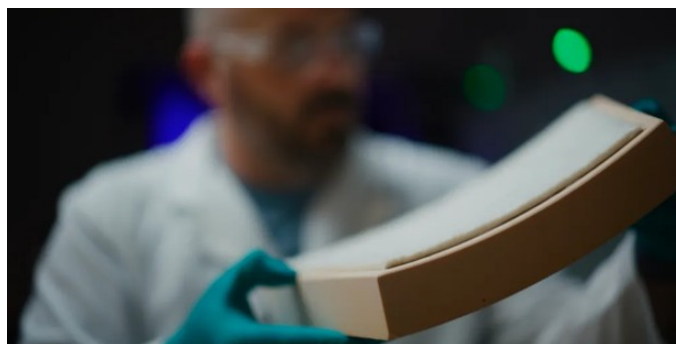
- **United Semiconductors (Orbital Semiconductor Manufacturing):** In November '25, Varda [entered a multi-flight joint development agreement](#) with United Semiconductors to produce semiconductor materials in microgravity for Earth-based use. United Semiconductors is a developer of in-space semiconductor manufacturing technology, having conducted testing on the ISS. Under the agreement, Varda would provide access to its W-Series reentry vehicles for United Semiconductor payloads.
- **Southern Launch (Reentry Operations):** In September 2025, Varda [extended a commercial agreement](#) with [Southern Launch](#) (Australian launch services provider) to support up to 20 reentries through the Koonibba Test Range in Australia, ensuring ground infrastructure for capsule recoveries and aligning orbital return operations through 2028.
- **Benchmark Space Systems (Reentry Support):** In October 2025, Benchmark Space Systems (propulsion technology provider) partnered with Varda to provide atmospheric reentry systems and propulsion technologies. Benchmark has publicly described how Varda's W-4 spacecraft featured five Benchmark 20 newton thrusters which enabled the vehicle's entry burn.
- **NASA (Technology Support / Materials):** Varda is working with NASA through a 2023 Tipping Point award to produce C-PICA and conduct flight testing through NASA's Flight Opportunities program. The agency also provides technical support in setting up manufacturing and collecting flight data. Varda was the first company to license C-PICA, a novel heat shield material developed by NASA designed to withstand the extreme temperatures of atmospheric reentry with 30-50% mass savings compared to PICA, an older heat shield technology.
- **LeoLabs & Anduril (Orbital Tracking and AI Integration):** In September 2025, [Varda collaborated with LeoLabs and defense technology firm Anduril](#) to demonstrate real-time tracking and reporting of orbital maneuvers ahead of a hypersonic reentry using LeoLabs' Global Radar Network and Anduril's Lattice software. The joint demonstration, centered on Varda's W-3 capsule operations, highlighted integration of commercial space domain awareness data with resilient, low-latency analytics and explored future use of deployable Scout radar systems for hypersonic reentry monitoring.
- **SSPC (Microgravity Crystallization Research):** In September 2024, Varda [announced a research collaboration](#) with the Research Ireland Centre for Pharmaceuticals (SSPC) to advance the mathematical modeling of crystallization in microgravity, focusing on how gravity influences polymorphic outcomes relevant to pharmaceutical manufacturing.

**Exhibit 13:** 5 Benchmark 20-Newton Thrusters Used to Initiate Reentry of W-Series Spacecraft



Source: Benchmark Space Systems

**Exhibit 14:** C-PICA Heatshield Technology Handled by an Engineer



Source: NASA

### Notable Potential Competitors

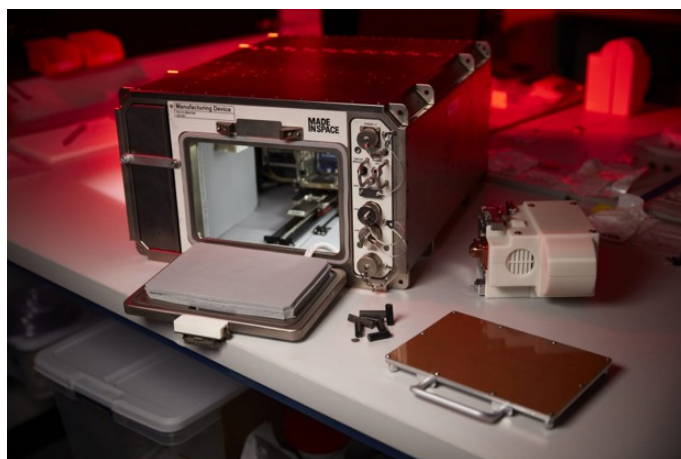
- Space Forge (Private):** [Space Forge](#) is a UK-based space manufacturing startup founded in 2018 building reusable orbital platforms (ForgeStar series) to produce value-added materials and advanced compounds in microgravity and return them to Earth. Its ForgeStar satellites are designed to enable microgravity manufacturing plus re-entry logistics, closely overlapping Varda's model of off-planet production with Earth return. Both companies aim to commercialize space-enabled industrial processes, though Space Forge emphasizes modular satellite factories and materials such as semiconductors and super-materials as opposed to pharmaceutical manufacturing. According to PitchBook, the company has raised \$39mn to-date with investors including NATO Innovation Fund, the European Space Agency, and Intel, among others.
- Redwire (NYSE: RDW):** [Redwire](#) is an American aerospace manufacturer and space infrastructure tech company that develops deployable structures for space systems, payload integration, sensors, and other critical space infrastructure systems. Its portfolio (particularly with the [acquisition of Made In Space](#)) overlaps with Varda's manufacturing niche in that Redwire provides microgravity production and assembly infrastructure though Redwire's focus is more on enabling and integrating manufacturing hardware than operating independent orbital factories.
- Voyager (NYSE: VOYG - Covered by Kristine Liwag):** [Voyager](#) is a defense and space tech company building broad space infrastructure, including commercial space stations like the next-gen Starlab, payload services, and microgravity research environments. While not specifically focused on product manufacturing and return like Varda, Voyager's orbital platforms and microgravity research infrastructure could overlap by hosting experiments or manufacturing payloads that align with Varda's service customers. We note that recently, [Starlab and space-research startup, Helogen](#) announced a partnership to integrate Helogen's autonomous biomedical operating systems into the Starlab ecosystem to enable scalable life sciences research and biomanufacturing onboard the commercial



station, potentially broadening the range of microgravity research and production hosted on Voyager's infrastructure.

- **Vast (Private):** Vast is a space habitat and station company developing commercial orbital platforms designed for research, manufacturing, and long-duration human operations. Its microgravity labs aboard these stations could serve in-orbit manufacturing and R&D activities that align with Varda's target markets, though Vast's primary mission is building habitats and platforms rather than dedicated manufacturing vehicles with Earth return capability.

**Exhibit 15:** Redwire Additive Manufacturing Hardware Launched to ISS



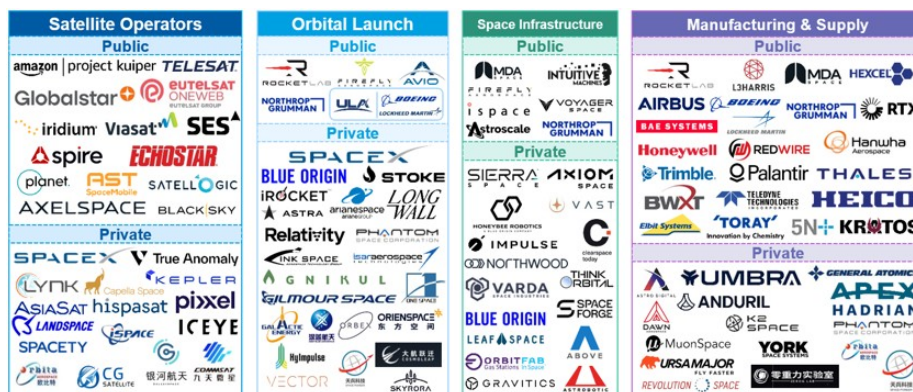
Source: Redwire

**Exhibit 16:** Space Forge ForgeStar Satellite Concept for Semiconductor and Alloy Manufacturing



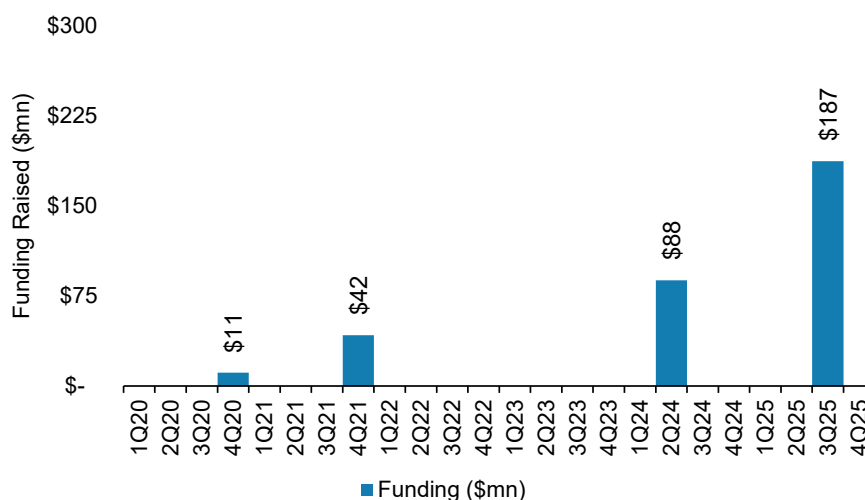
Source: Space Forge

**Exhibit 17:**  
Global Space Enablers



Source: Morgan Stanley Research

**Investors & Funding.** According to PitchBook, Varda has raised \$328mn to-date (valuation not reported) with its most recent funding in July 2025 where it raised a \$187mn Series C round led by Natural Capital and Shrug Capital. Other investors include Founders Fund, Peter Thiel, Khosla Ventures, Caffeinated Capital, and General Catalyst, among others.

**Exhibit 18:** Varda Space Historical Funding Rounds (Sourced from PitchBook)

Source: PitchBook, Morgan Stanley Research

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#### Timeline of Major Developments:

- **2021:** Varda is founded and exits stealth with seed funding from Founders Fund and Lux.
- **October 2021:** Raised \$42mn in a Series A round led by Khosla Ventures and Caffeinated Capital.
- **February 2024:** Completed W-1 mission, returning Form III crystals of HIV drug, Ritonavir, after manufacturing them in orbit.
- **April 2024:** Raised \$88mn in a Series B round led by Caffeinated Capital with participation from Valor Equity Partners, Founders Fund, Lux Capital, Khosla Ventures, General Catalyst, and others.
- **January-May 2025:** Launched and returned multiple W-series missions (W-2 and W-3).
- **July 2025:** Raised \$187mn in a Series C round led by Shrug Capital and Natural Capital with participation from Founders Fund, Lux Capital, Caffeinated Capital, Khosla Ventures, Peter Thiel, and others.
- **January 2026:** The W-5 capsule completed a successful reentry and recovery in South Australia, marking the first use of Varda's vertically integrated satellite bus.

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|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1520              | 41%        | 425                              | 48%            | 28%                  | 685                                               | 42%                   |
| Equal-weight/Hold     | 1579              | 43%        | 375                              | 42%            | 24%                  | 713                                               | 44%                   |
| Not-Rated/Hold        | 4                 | 0%         | 1                                | 0%             | 25%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 592               | 16%        | 89                               | 10%            | 15%                  | 232                                               | 14%                   |
| Total                 | 3,695             |            | 890                              |                |                      | 1631                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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